NOT MEASUREMENT SENSITIVE

MIL-S-20708/66E
13 March 1989

SUPERSEDING
MIL-S-20708/66D
30 July 1987

MILITARY SPECIFICATION SHEET

SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 31TRX6A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708

TABLE I Requirements

| Requirement | Value | Unit | Tolerance |
|---|---|---|---|
| Frequency | 60 | Hz | ±1% |
| Primary Voltage | 115 | volts | ±1% |
| Primary Current | 440 0 | milliamps | maximum |
| Primary Power | 6 6 | watts | maximum |
| Impedance | | | |
| Zss | 24 0-33 0 | ohms | min -max |
| Impedance Angle | | | |
| Zss | 16 0-24 0 | degrees | min -max |
| Transformation Ratio | 0 783 | ----- | ±2% |
| Phase Shift (Lead) | 6 5 | degrees | maximum |
| Electrical Error | 10 0 | minutes | maximum |
| Receiver Error | 36 0 | minutes | maximum |
| Null Voltage | | | |
| Total | 170 0 | millivolts | maximum |
| Fundamental | 35 0 | millivolts | maximum |
| Synchronizing Time 30° ±2° | 1 0 | seconds | maximum |
| Synchronizing Time 177° ±2° | 2 0 | seconds | maximum |
| Torque Gradient | 0 40 | ounce-inches per degree | minimum |
| Radial Play | 0 0008 | inches | maximum |
| End Play | 0 0030 | inches | ±0 0020 |
| Temperature Rise | 35 0 | degrees (C) | maximum |
| Variation of Voltage (+10%) and Freq (-5%) | 9 4 | watts | maximum |

This unit shall not be subjected to the altitude requirements of MIL-S-20708

(E) denotes changes


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TABLE II Military part number variant characteristics

| Military Part No 1/ | A ± 010 | B ± 010 | C Maximum | Shaft Style | Terminal End | U + 0000 − 0002 |
|---|---|---|---|---|---|---|
| M20708/66-01A | 700 | 730 | 3 380 | Splined | Terminal Board | 2405 |

1/Military part number M20708/66-01A shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

| SUPERSEDED MILITARY PART NUMBERS | | NEW MILITARY PART NUMBERS |
|---|---|---|
| MIL-S-20708/66C | MIL-S-20708/66B | |
| M20708/66C-001 | M20708/66-001 | M20708/66-01A |

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians
Army-AR
Navy-AS
Air Force-85

Review Activities
Army-MI,AV
Navy-OS
Air Force-11,17,99
DLA-ES

User Activities
Navy-MC CG

Preparing Activity
Navy-AS

(Project 5990-0349-40)

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